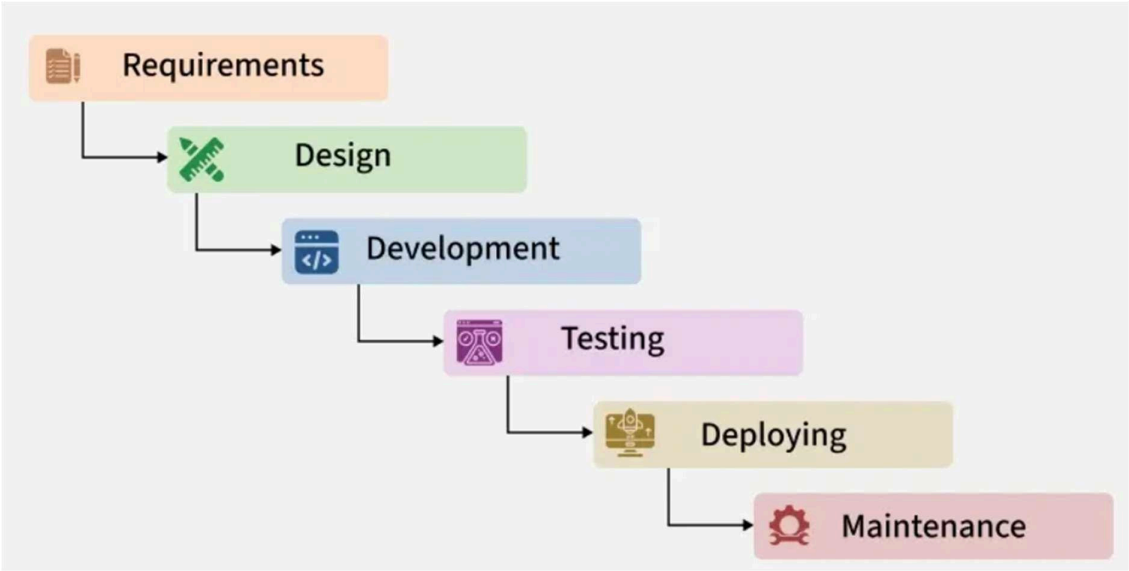
Software Process

1. Waterfall Model

Waterfall model provides a systematic, sequential approach to software development. The waterfall model is also called a **linear sequential model**.

The software development starts with a **requirements gathering phase**. Then progresses through:

1. Analysis
2. Design
3. Coding
4. Testing
5. Maintenance



Waterfall Model

Phases of Waterfall Model

1. Requirements Gathering and Analysis

In this phase, the basic requirements of the system must be understood by the software engineer. All these requirements are documented for further use.

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2. Design

In this phase, the requirements are translated into easier format, using which coding can be done efficiently.

The design focuses on program attributes such as:

- Data structure
- Software architecture educational
- Interface representation
- Algorithmic details

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3. Coding

The next phase is coding, in which the design is translated into machine-readable form. Programs are created in this phase.

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4. Testing

The purpose of testing is to uncover errors, fix the bugs and meet the customer requirements. Monal institute the testing ensures execution of all the paths and functional behaviours.

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5. Maintenance

Maintenance is the longest life cycle phase. When the system is installed and put into practical use, some errors may get introduced. Correcting such errors is the major purpose of maintenance activity.

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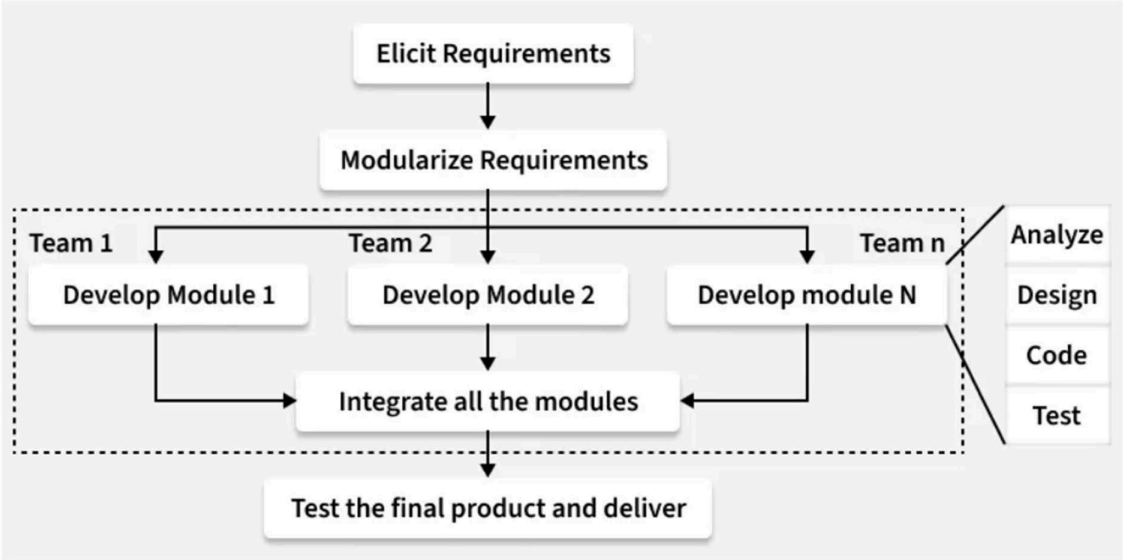
2. <u>RAD Model</u>

The Rapid Application Development (RAD) model is a type of incremental process model in which there is an extremely short development cycle.

Using the RAD model, the fully functional software system can be developed within **60 to 90 days**.

The various phases in RAD model are:

1. Requirements gathering
2. Analysis and planning
3. Design
4. Build
5. Deployment



RAD Model

Phases of RAD Model

1. Requirements Gathering

In this phase, the basic requirements of the system must be understood by the software engineer.

All these requirements are documented for further use.

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2. Analysis and Planning

During the analysis and planning phase, the analysis on the gathered requirements is made.

Planning for various software development activities is done.

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3. Design

During the design phase, various models such as:

- Business model
- Data model
- Process model

are created.

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4. Build

During the build phase, the implementation code for the software system is created. The functionalities developed by all the teams are integrated to form a whole system.

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5. Deployment

Finally, the software system created by integrating all the software components, is deployed in the real world.

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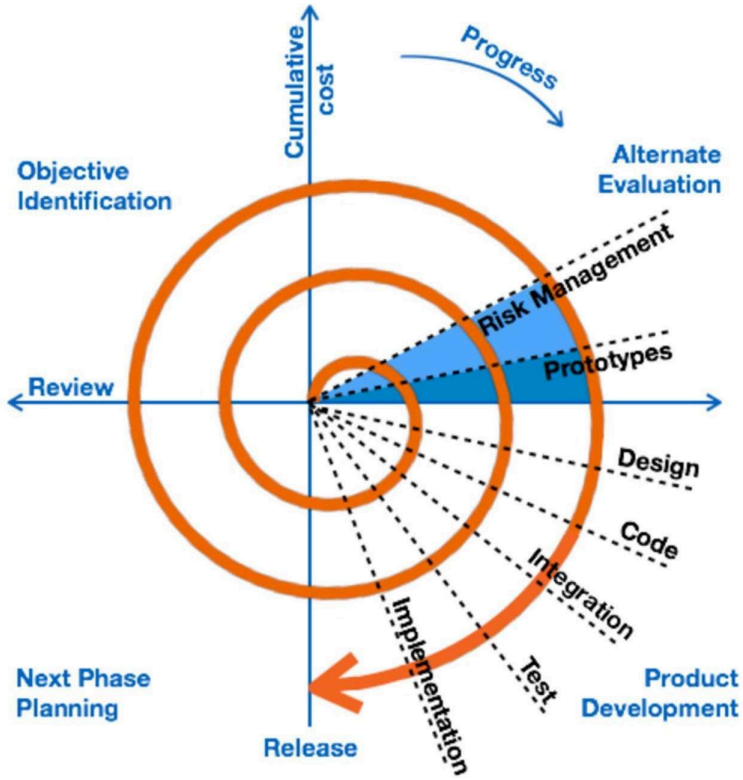
3. Spiral Model

Spiral model is a type of evolutionary process model. It possesses the iterative nature of prototyping model and the systematic approaches of the linear sequential model.

The spiral model is divided into a number of framework activities, which are denoted by task regions. Usually, there are six task regions in the spiral model.

They are:

1. Customer communication
2. Planning
3. Risk analysis
4. Engineering
5. Construct and Release
6. Customer evaluation



Task Regions of Spiral Model

1. Customer Communication

In this region, customer communication about the project is established.

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2. Planning

In this region, the planning is done in order to define the resources timeline and other project related activities.

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3. Risk Analysis

The tasks required to calculate the technical and management risks are carried out in this region.

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4. Engineering

In this task region, the tasks required to build one or more representations of the applications are carried out.

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5. Construct and Release

All the tasks required to construct, test and install the application are carried out in this task region.

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6. Customer Evaluation

In this region, the customer feedback is obtained. Based on the feedback, the required tasks are implemented at the installation stage.

Agile Process

The agile process represents a methodology in project management that emphasizes **flexibility, collaboration,** and **continuous improvement**.

It empowers development teams to adapt swiftly to evolving requirements while ensuring the delivery of value to stakeholders.

This approach is characterized by its **iterative nature**.

Example: Extreme Programming (XP)

The Agile Focuses on

Individuals and interactions	Over	Process and tools
Working software	Over	Comprehensive documentation
Customer collaboration	Over	Contact negotiation
Responding to change	Over	Following a plan

The aim of the agile process is to deliver the working model of software quickly to the customer.

The 12 Agile Principles

- Prioritize stakeholder satisfaction through the early and continuous delivery of functional software.
- Actively accommodate shifting requirements throughout the development cycle.
- Deliver working software frequently within abbreviated timeframes.
- Ensure consistent collaboration between business stakeholders and developers.
- Provide motivation and technical support to the project team.
- Utilize functional software as the primary metric for tracking progress.
- Maintain a sustainable and constant development pace.
- Foster technical excellence to enhance overall agility.
- Dedicate attention to superior design and technical quality.
- Uphold simplicity as an essential element of the development process.
- Empower self-organizing teams to produce optimal architecture and designs.
- Regularly evaluate and adjust team behavior to maximize operational effectiveness.

Extreme Programming (XP)

The Agile Process is a project management approach specifically designed for software development that prioritizes:

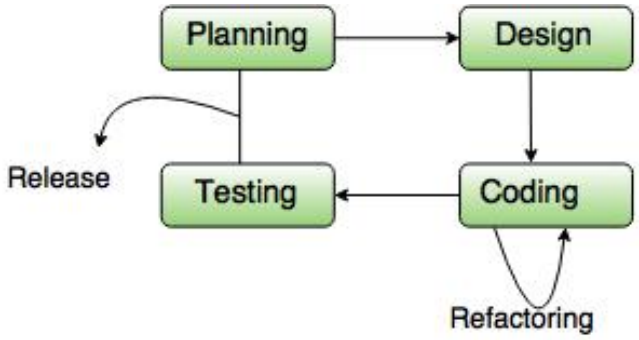
- **Flexibility:** The ability to adapt to changes quickly rather than sticking to a rigid, fixed plan.
- **Collaboration:** A people-centric approach involving constant communication between developers and customers.
- **Continuous Improvement:** The project evolves and improves through iterative changes.

Core Characteristics

- **Iterative Method:** It follows a repetitive cycle where the software is developed in small increments rather than all at once.
- **Adaptability:** If a problem arises or requirements change, the team can immediately adjust the plan to recover and maintain progress.
- **Value Delivery:** By being responsive to change, the process ensures that the final product delivers the highest possible value to the customer.

Extreme Programming (XP)

Extreme Programming is recognized as a prominent **Agile process model**. It follows an iterative and incremental approach, specifically designed to address evolving requirements and deliver high value to stakeholders.



The XP Process Flow

The XP workflow operates through a continuous cycle consisting of the following phases:

1. **Planning:** Customers define requirements, which are captured as Story Cards in collaboration with developers.
2. **Design:** Story cards are decomposed into small tasks, and resource estimations are established.

3. **Coding:** Developers implement the code, applying refactoring as necessary to refine the system structure.
4. **Testing:** The system is evaluated with the customer; feedback triggers new story cards and cycle repetition.
5. **Release:** The functional software is delivered through frequent, small-scale releases.

The 5 Core Values of XP

The XP methodology is driven by five essential values:

Value	Description
Communication	Establishing an effective exchange between developers and all stakeholders.
Simplicity	Prioritizing simple designs to facilitate future modifications.
Feedback	Continuous input from customers and the team to drive improvement.
Courage	Readiness to recognize new requirements and adjust plans accordingly.
Respect	Earning stakeholder trust by adhering to XP values and delivering quality.